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import numpy as np
import matplotlib.pyplot as plt

# Mean Square Error calculation
def compute_error_for_line_given_points(b, m, points):
    totalError = 0
    for i in range(len(points)):
        x = points[i, 0]
        y = points[i, 1]
        totalError += (y - (m * x + b)) ** 2
    return totalError / float(len(points))

# Compute gradients and update for one step
def step_gradient(b_current, m_current, points, learning_rate):
    b_gradient = 0
    m_gradient = 0
    N = float(len(points))
    for i in range(len(points)):
        x = points[i, 0]
        y = points[i, 1]
        b_gradient += -(2 / N) * (y - ((m_current * x) + b_current))
        m_gradient += -(2 / N) * x * (y - ((m_current * x) + b_current))
    new_b = b_current - (learning_rate * b_gradient)
    new_m = m_current - (learning_rate * m_gradient)
    return new_b, new_m

# Batch Gradient Descent
def batch_gradient_descent(points, initial_b, initial_m, learning_rate, num_iterations):
    b = initial_b
    m = initial_m
    for i in range(num_iterations):
        b, m = step_gradient(b, m, points, learning_rate)
    return b, m

# Stochastic Gradient Descent (SGD)
def stochastic_gradient_descent(points, initial_b, initial_m, learning_rate, num_iterations):
    b = initial_b
    m = initial_m
    for i in range(num_iterations):
        for point in points: # Update parameters for each point
            x, y = point[0], point[1]
            b_gradient = -2 * (y - (m * x + b))
            m_gradient = -2 * x * (y - (m * x + b))
            b -= learning_rate * b_gradient
            m -= learning_rate * m_gradient
    return b, m

# Mini-Batch Gradient Descent
def mini_batch_gradient_descent(points, initial_b, initial_m, learning_rate, num_iterations):
    b = initial_b
    m = initial_m
    for i in range(num_iterations):
        np.random.shuffle(points) # Shuffle the dataset for mini-batches
        for j in range(0, len(points), batch_size):
            mini_batch = points[j:j + batch_size]
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        b, m = step_gradient(b, m, mini_batch, learning_rate)
    return b, m

# Plot results
def plot_results(points, b, m, method_name):
    x = points[:, 0]
    y = points[:, 1]

    plt.scatter(x, y, color='blue', label='Data Points')
    plt.plot(x, m * x + b, color='red', label=f'Regression Line ({method_name})')
    plt.title(f'Results of {method_name}')
    plt.xlabel('X')
    plt.ylabel('Y')
    plt.legend()
    plt.show()

# Run function to test all methods
def run():
    # Load dataset
    points = np.genfromtxt("data.csv", delimiter=",")

    # Hyperparameters
    learning_rate = 0.01
    initial_b = 0
    initial_m = 0
    num_iterations = 1000
    batch_size = 20 # For mini-batch gradient descent

    # Batch Gradient Descent
    print("Running Batch Gradient Descent...")
    b_batch, m_batch = batch_gradient_descent(points, initial_b, initial_m, learning_rate,
    batch_error = compute_error_for_line_given_points(b_batch, m_batch, points)
    print(f"Batch Gradient Descent: b = {b_batch}, m = {m_batch}, MSE = {batch_error}")
    plot_results(points, b_batch, m_batch, "Batch Gradient Descent")

    # Stochastic Gradient Descent
    print("\nRunning Stochastic Gradient Descent...")
    b_sgd, m_sgd = stochastic_gradient_descent(points, initial_b, initial_m, learning_rate,
    sgd_error = compute_error_for_line_given_points(b_sgd, m_sgd, points)
    print(f"Stochastic Gradient Descent: b = {b_sgd}, m = {m_sgd}, MSE = {sgd_error}")
    plot_results(points, b_sgd, m_sgd, "Stochastic Gradient Descent")

    # Mini-Batch Gradient Descent
    print("\nRunning Mini-Batch Gradient Descent...")
    b_mini, m_mini = mini_batch_gradient_descent(points, initial_b, initial_m, learning_rate,
    mini_error = compute_error_for_line_given_points(b_mini, m_mini, points)
    print(f"Mini-Batch Gradient Descent: b = {b_mini}, m = {m_mini}, MSE = {mini_error}")
    plot_results(points, b_mini, m_mini, "Mini-Batch Gradient Descent")

if __name__ == '__main__':
    run()

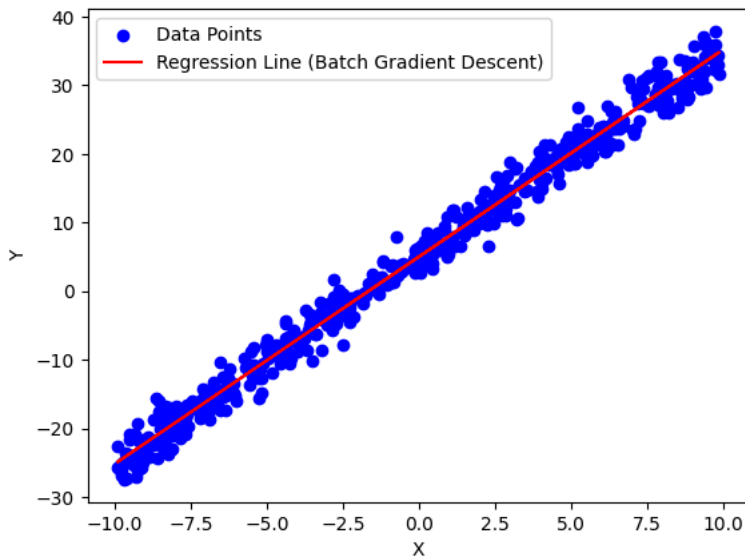
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Running Batch Gradient Descent...

Batch Gradient Descent: $b = 5.015901321407632$, $m = 3.023000462282122$, $MSE = 4.02974248$

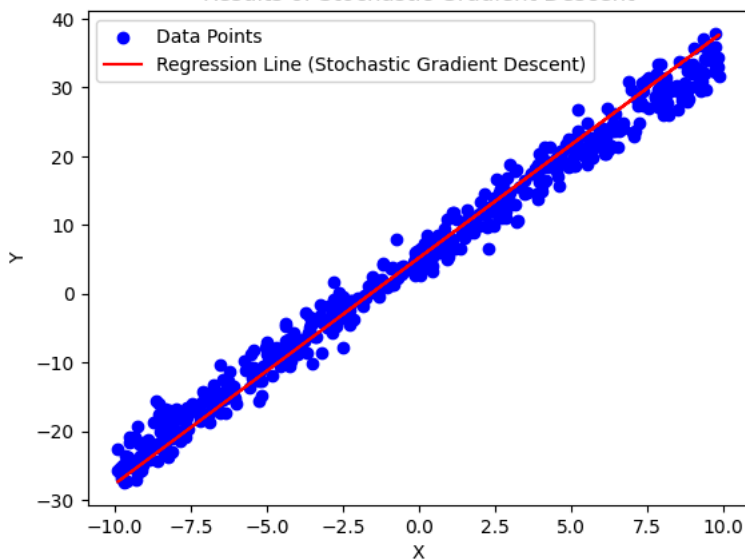
Results of Batch Gradient Descent



Running Stochastic Gradient Descent...

Stochastic Gradient Descent: $b = 5.211877752563265$, $m = 3.2933747273223397$, $MSE = 6.66$

Results of Stochastic Gradient Descent



Running Mini-Batch Gradient Descent...

Mini-Batch Gradient Descent: $b = 5.02380765468742$, $m = 3.0399454318809647$, $MSE = 4.046$

Results of Mini-Batch Gradient Descent

